
RNN-Based Realized Volatility Forecasting for NVIDIA

An empirical comparison of single-layer and multi-layer recurrent neural networks

Research Report

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Repository

github.com/ekuo29/rnn_realized_volatility_forecasting

Abstract

This study evaluates recurrent neural networks for forecasting NVIDIA Corporation's next-day realized volatility. Single-layer and two-layer SimpleRNN architectures are compared under univariate and multivariate feature specifications. Candidate predictors include log realized volatility, its log 10-day moving average, and log VIX. Each observation is represented by a 21-day sequence. Models are estimated using 2018-2022 data, selected with 2023-2024 validation performance, and evaluated once on the reserved 2025 test period. The tuning grid varies recurrent units, dropout, learning rate, and optimizer while using Huber loss and early stopping. The selected two-layer model uses log realized volatility and its 10-day moving average, achieves a validation RMSE of 0.006427, and records a test RMSE of 0.006548. Its validation advantage over the tuned single-layer model is approximately 0.08 percent. The results indicate that NVIDIA's own recent volatility history contains most of the useful signal in the tested feature set, while additional recurrent depth provides only a marginal improvement.

Keywords: realized volatility; recurrent neural network; time series; NVIDIA; volatility forecasting

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1. Introduction

1.1 Motivation

Realized volatility is an observable, ex-post measure of the variation accumulated by an asset over a trading day. It is constructed from high-frequency intraday returns and is commonly used as an empirical approximation of integrated volatility. Integrated volatility, by contrast, is a latent continuous-time quantity and cannot be observed directly.

Accurate realized-volatility forecasts support portfolio allocation, volatility targeting, market-risk measurement, derivatives pricing, and hedging. Volatility also displays persistence and clustering: periods of elevated volatility are frequently followed by continued instability, while calm periods tend to persist. This temporal dependence motivates a sequential forecasting model.

NVIDIA is selected because its volatility varies substantially over time and contains pronounced episodes of market stress. The analysis focuses exclusively on the RNN implementation and does not report results from other model classes.

1.2 Research objective

The objective is to identify the RNN specification that minimizes one-day-ahead forecast error while preserving a strict chronological evaluation design. The study addresses three questions:

1. Does a two-layer RNN improve validation performance relative to a single-layer RNN?
2. Do moving-average volatility and VIX add predictive information beyond NVIDIA's recent realized volatility?
3. Does the validation-selected model maintain comparable accuracy on the unseen 2025 test period?

1.3 Research contribution

The study provides a transparent RNN-only forecasting workflow that separates model development, hyperparameter selection, and final testing. Feature transformations are fitted with training observations only, candidate specifications are compared using validation data, and the test period is reserved for a single final evaluation.

2. Data and Research Design

2.1 Data and variables

Daily realized-volatility observations are loaded from the provided `daily_metrics.pkl` dataset and filtered for NVIDIA. Dates are converted to a chronological index. Daily VIX observations are aligned by date and joined to the NVIDIA series. The resulting sample spans 2018 through 2025.

Variable	Definition	Role
log_rv	Log of daily NVIDIA realized volatility	Primary state variable
log_rv_ma10	Log of the 10-day moving average of realized volatility	Smoothed volatility state
log_vix	Log of the CBOE Volatility Index	Market-wide uncertainty proxy
target	One-day-ahead log realized volatility	Forecasting outcome

2.2 Chronological sample partition

Period	Sample	Purpose
2018-2022	Training	Estimate data transformations and model parameters
2023-2024	Validation	Compare features, architectures, and hyperparameters
2025	Test	Evaluate the selected model once

2.3 Preprocessing and forecast target

Feature variables and the forecasting target are standardized separately using means and standard deviations estimated from the training sample. The fitted transformations are then applied to validation and test observations. This procedure prevents future information from entering model estimation.

Each sample is converted into overlapping sequences of 21 observations. The target is next-day log realized volatility:

$$y_t = \log(RV_{t+1})$$

Model predictions are inverse-scaled and exponentiated before error metrics are calculated on the original realized-volatility scale.

3. Model Implementation

3.1 Architecture

The implementation compares a single-layer SimpleRNN with a two-layer stacked SimpleRNN. Both architectures use tanh activation, dropout regularization, a one-neuron dense output layer, Huber loss, and early stopping.

Component	Single-layer RNN	Multi-layer RNN
Input	21 observations by selected features	21 observations by selected features
Recurrent structure	One SimpleRNN layer	Two stacked SimpleRNN layers
Regularization	Dropout after recurrent layer	Dropout after each recurrent layer
Output	Dense(1)	Dense(1)
Loss	Huber	Huber

3.2 Feature and model development

Development begins with a fixed 21-day sequence length. Within each architecture, four feature specifications are compared: log realized volatility alone; log realized volatility with its 10-day moving average; log realized volatility with VIX; and a three-variable specification containing all candidate predictors. The strongest feature specification is then tuned.

3.3 Hyperparameter search

Hyperparameter	Candidate values
Recurrent units	32, 64
Dropout	0.1, 0.2
Learning rate	0.0005, 0.001, 0.002
Optimizer	Adam, RMSprop, SGD
Training horizon	Maximum 100 epochs; batch size 32; patience 8

Validation RMSE is the primary model-selection criterion:

$$RMSE = \sqrt{(1/n) \sum_{i=1}^n (RV_i - RVhat_i)^2}$$

Mean absolute error is retained as a secondary diagnostic. Early stopping monitors validation loss with a patience of eight epochs and restores the best observed weights.

4. Empirical Results

4.1 Development-stage performance

Table 4 reports validation performance for the untuned architecture and feature comparisons.

Architecture	Feature specification	Validation RMSE	Validation MAE
Single-layer	log_rv	0.006473	0.004672
Single-layer	log_rv + log_rv_ma10	0.006441	0.004656
Single-layer	log_rv + log_vix	0.006677	0.004784
Single-layer	log_rv + log_rv_ma10 + log_vix	0.006564	0.004701
Multi-layer	log_rv	0.006564	0.004740
Multi-layer	log_rv + log_rv_ma10	0.006542	0.004691
Multi-layer	log_rv + log_vix	0.006805	0.004846
Multi-layer	log_rv + log_rv_ma10 + log_vix	0.006609	0.004804

The development-stage results yield four main observations:

- Adding the 10-day moving average improves validation RMSE in both architectures.
- Adding VIX without the moving average worsens performance relative to the univariate specification.
- The three-variable specification does not outperform the two-variable realized-volatility specification.
- The strongest untuned model in each architecture uses log_rv and log_rv_ma10.

4.2 Interpretation of feature selection

The results suggest that the moving average provides useful information about the persistent component of NVIDIA's volatility. VIX is economically motivated as a market-wide uncertainty proxy, but it does not add sufficient incremental predictive information in the tested sample. This is a sample-specific result and should not be interpreted as evidence that VIX is generally uninformative.

4. Empirical Results - Continued

4.3 Tuned-model selection

Candidate	Features	Units	Dropout	Learning rate	Optimizer	Validation RMSE
Tuned single-layer	log_rv, log_rv_ma10	64	0.2	0.0020	RMSprop	0.006432
Tuned multi-layer	log_rv, log_rv_ma10	32	0.2	0.0005	Adam	0.006427

The tuned two-layer model achieves the lowest validation RMSE. Its absolute improvement over the tuned single-layer model is 0.000005, equivalent to approximately 0.08 percent. The result supports selection of the multi-layer model but does not indicate a large practical advantage from additional depth.

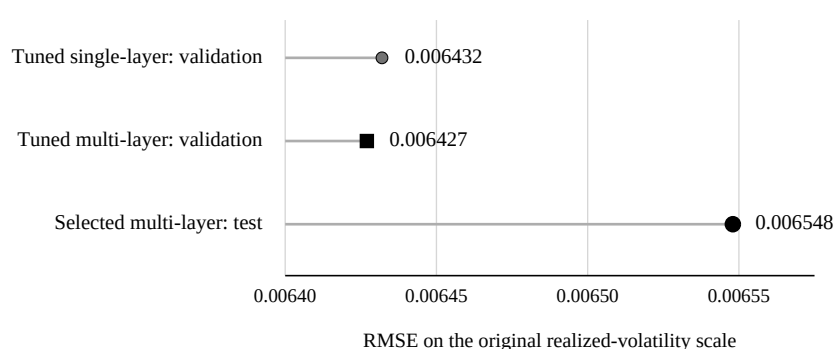


Figure 1. RMSE comparison for the two validation finalists and the selected model's 2025 test performance. The narrow axis highlights the small validation difference; comparisons should rely on the printed values.

4.4 Out-of-sample performance

Selected model	Validation RMSE	Validation MAE	Test period	Test RMSE
Tuned multi-layer RNN	0.006427	0.004620	2025	0.006548

Test RMSE is approximately 1.9 percent higher than validation RMSE. The relatively small increase indicates limited performance deterioration between model selection and final evaluation. This comparison does not, by itself, establish statistical significance or economic value.

5. Diagnostic Analysis

5.1 Validation diagnostics

The development plots show training and validation loss together with validation forecasts for a representative multi-layer candidate. Loss stabilizes after the early epochs, supporting the use of early stopping. The forecast follows the general level and persistence of realized volatility but smooths isolated spikes.

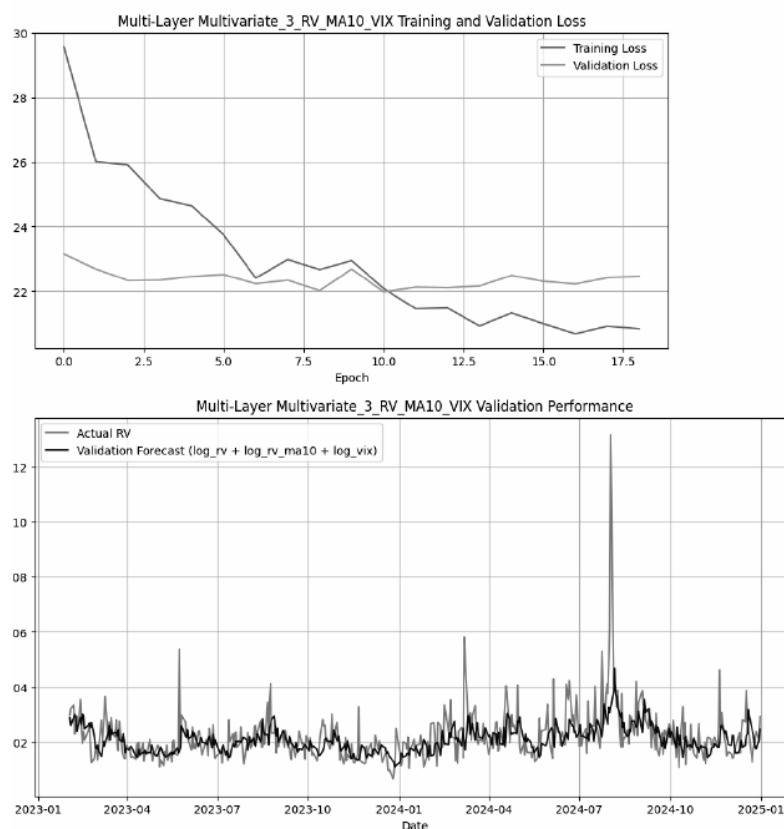


Figure 2. Training and validation loss, followed by validation forecasts for the representative multi-layer multivariate RNN. Source: output retained from the completed RNN analysis; figure converted to grayscale.

5. Diagnostic Analysis - Continued

5.2 Test-period forecast

Figure 3 compares actual realized volatility with the selected model's 2025 forecasts. The forecast tracks persistent movements and the broad volatility level. The largest realized-volatility spike is substantially underestimated, illustrating the difficulty of predicting abrupt market shocks from lagged observations.

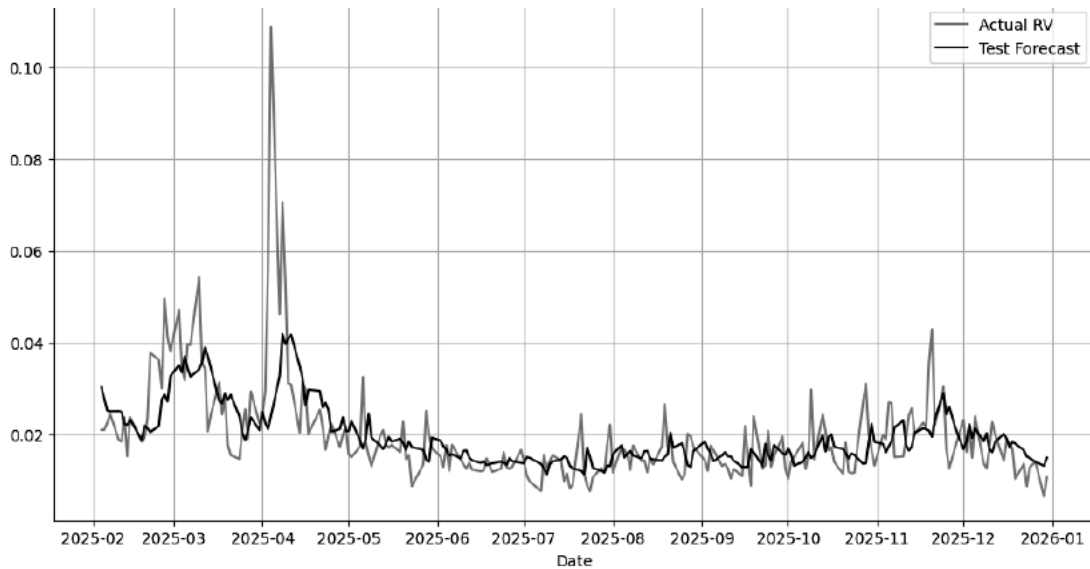


Figure 3. Actual and forecast NVIDIA realized volatility during the 2025 test period. Source: output retained from the completed RNN analysis; figure converted to grayscale.

The diagnostic evidence supports three conclusions:

- The model captures persistent movements more effectively than isolated shocks.
- The forecast series is smoother than realized volatility, particularly during extreme observations.
- Test error remains close to validation error despite the presence of a large test-period volatility spike.

6. Discussion and Limitations

6.1 Discussion

The findings indicate that NVIDIA's recent volatility history is the principal source of predictive information in the tested feature set. The 10-day moving average contributes a smoothed volatility state, while VIX is excluded from the winning specification. The two-layer architecture slightly outperforms the single-layer alternative, but the 0.08 percent relative improvement is too small to support a broad claim that greater recurrent depth is economically important.

The central result is therefore one of parsimony. Sequential structure and feature construction matter, while added depth provides only an incremental gain. A single-layer RNN may remain attractive when interpretability, computational cost, or implementation simplicity is important.

6.2 Limitations

- The analysis focuses on one asset and may not generalize to other securities.
- The final test period contains only 2025 observations.
- RMSE emphasizes large errors and does not fully characterize performance during extreme events.
- Neural-network estimates may vary slightly across runs because of random initialization.
- The RNN-only implementation does not include an external statistical forecasting benchmark.
- Economic value is not evaluated through a portfolio, hedging, or trading application.

7. Conclusion

A two-layer SimpleRNN using log realized volatility and its log 10-day moving average achieves the best validation performance among the tested RNN specifications. The selected model records a validation RMSE of 0.006427 and a 2025 test RMSE of 0.006548. Its marginal advantage over the tuned single-layer model indicates that recent volatility history and disciplined feature construction are more consequential than additional recurrent depth.

The study supports a chronological modeling process in which preprocessing parameters are estimated from training data only, candidate models are selected using validation performance, and the test sample is reserved for a single final evaluation.

References

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